Robust Adaptive Control of Flexible Joint Robots with Joint Torque Feedback*

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Abstract

This paper addresses motion control issues of flexible joint robot manipulators using joint torque feedback. A two-stage control scheme consisting of a motion controller and a joint torque controller is established in a systematic way for the general n-link case. To deal with uncertainties in the robotic system, a robust adaptive control algorithm is developed assuming that all system parameters including the joint flexibility values are unknown except for some of their bounds. The system stability is analyzed via Lyapunov stability theory. The result has the distinct feature that no restriction of joint flexibility, nor exact knowledge of the parameters of rotor subsystem is required. The outcome is a useful framework for generalization of control methods previously developed for rigid robots.

Simulation results verify the effectiveness of the proposed control methods.

1 Introduction

With the advent of robot technology, the use of joint torque measurement for the control of robot manipulators has been proposed by several authors [1—4]. However, concerning the control of flexible joint robot, control schemes using joint torque feedback techniques are still few, and most of them are limited to applications of one single joint [2, 4]. Recent works [4, 7] have pointed out that joint flexibility causes difficulties in modeling and control. Control of robots based on rigid-body dynamics formulation is inadequate in dealing with more stringent operating conditions. In some cases, joint flexibility can even lead to instability if it is neglected in the control design. Much effort has been devoted to the control of flexible joint robots. Dynamic models and control methods based on different techniques have been proposed in literatures. The main methods are feedback linearization, singular perturbation, integral manifold and adaptive techniques. Marino and Nicosia [5] applied a singular perturbation approach to design controllers for manipulators with flexible joints. Spong [6, 7] introduced two nonlinear control techniques, namely, the feedback linearization technique and the integral manifold technique. Some adaptive control strategies have been proposed to deal with the case of parametric uncertainty in flexible joint robots. Utilizing the concept of integral manifolds, Khorasani [9] developed the corrected adaptive control laws. Relevant work was also conducted actively by Spong et al. [8]. Most control schemes using singular perturbation techniques are obtained under the assumption of weak joint flexibility (high rigidity). Chen and Fu [10] proposed an adaptive control scheme with a two-stage control algorithm, which does not require the restriction of weak joint flexibility but assumes the exact knowledge of parameters of the rotor subsystem. In work [11], an adaptive force control scheme for a single-link mechanism with joint flexibility was developed. However, the method can not be easily extended to the general n-link case.

In this paper, we consider motion control problem of flexible joint robots by using joint torque feedback. Based on the idea of two-stage control strategy, a control scheme consisting of a motion controller and a joint torque controller is established in a systematic way for the general n-link case. To deal with uncertainties in the robotic system, a robust adaptive control algorithm is developed assuming that all the system parameters are unknown except for some of their bounds. The system stability is analyzed via Lyapunov stability theory. The result compared with the previous work has the distinct feature that no restriction of weak joint flexibility, nor exact knowledge of the parameters of rotor subsystem is required. The outcome is a useful framework for generalization of control methods previously developed for rigid robots.

In Section 2, the control scheme with known parameters is presented. Section 3 is devoted to the robust adaptive control algorithm. Simulation results are illustrated in Section 4 to verify the effectiveness of the proposed control methods. And conclusions are given in Section 5.

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2 Control scheme with all parameters known

Consider the dynamic equations of a flexible joint robot with joint torque measurements [6, 12]:

$$D(q_l)\ddot{q}_l + C(q_l,\dot{q}_l)\dot{q}_l + G(q_l) = \tau_s \quad (1)$$

$$I_m\ddot{q}_m + B_m\dot{q}_m + \tau_s = u \quad (2)$$

and

$$\tau_s = K_s(q_m - q_l) \quad (3)$$

where $D(q_l)$ is the $n \times n$ positive definite, symmetric inertia matrix; $C(q_l,\dot{q}_l)$ is the $n \times 1$ vector containing coriolis, centrifugal terms; $G(q_l)$ is the $n \times 1$ vector of gravitational forces; I_m is the $n \times n$ constant diagonal matrix with diagonal elements $\gamma_i(\gamma_i + 1)J_{mi}$, J_{mi} is the inertia of the rotor/gear; B_m is the $n \times n$ diagonal matrix of damping terms and K_s is a $n \times n$ diagonal matrix of the joint torsional stiffness. q_l and q_m are n dimensional vectors, represent the link angles and rotor angles respectively. u is the $n \times 1$ input torque control vector and τ_s is the vector of joint torque measurements.

The dynamic model has the following fundamental properties [13], which can be exploited to facilitate the control system design:

P1. The inertia matrix $D(q_l)$ is symmetric and positive definite.

P2. All of the constant parameters, such as link masses, moments of inertias, etc., appear as coefficients of known functions of the generalized coordinates.

P3. Using the Christoffel symbols, the matrix $\dot{D}(q_l) - 2C(q_l,\dot{q}_l)$ is a skew symmetric matrix.

For control design with known parameter case, the following assumptions are made:

A1 the signals q_l, $\dot{q}_l$, τ_s and $\dot{\tau}_s$ are measured and available for feedback control;

A2 the desired trajectory $q_d(t) \in C^4$ is bounded;

A3 the robot dynamics is known, i.e. all parameters are known precisely.

From equation (3), we have

$$q_m = K_s^{-1}\tau_s + q_l \quad (4)$$

Substituting equality (4) into equation (2), using equation (1) and the fact P1, we get:

$$I_{ts}\ddot{\tau}_s + B_{ts}\dot{\tau}_s + h_{ts}(q_l,\dot{q}_l,\tau_s) = u \quad (5)$$

where

$$I_{ts} = I_m K_s^{-1}$$
$$B_{ts} = B_m K_s^{-1}$$
$$h_{ts} = (I_m D^{-1} + I)\tau_s + B_m\dot{q}_l - I_m D^{-1}C - I_m D^{-1}G$$

Equations (1) and (5) constitute a dynamic representation for the flexible joint robot with the variable (q_l, τ_s).

We notice that link dynamics (1) has exactly the same formulation as the rigid one if τ_s is viewed as the *"input control torque"*. Hence, in order to control the robot manipulator to track the desired trajectory, motion control methods developed for rigid robot control could be used to generate these *"input control torque"* which we will denote here as *"desired joint torque"* τ_{sd} since joint flexibility exists. The next control objective is, naturally, to generate a suitable control torque u so that joint torque τ_s can follow the desired joint torque τ_{sd}, and thus q_l will follow the desired trajectory q_d. Based on this observation, a two stage control scheme is developed.

2.1. Motion controller

Let us define tracking errors:

$$e_l = q_l - q_d$$
$$e_t = \tau_s - \tau_{sd}$$

and denote

$$v_l = \dot{q}_d - \Lambda(q_l - q_d) = \dot{q}_d - \Lambda e_l$$

The desired joint torque τ_{sd} is generated by using the Slotine and Li method [14],

$$\tau_{sd} = D(q_l)\dot{v}_l + C(q_l,\dot{q}_l)v_l + G(q_l) - K_{lD}(\dot{q}_l - v_l) \quad (6)$$

where K_{lD} and Λ are diagonal matrices of positive gains.

Substituting (6) into (1) and defining $r_l = \dot{q}_l - v_l$ leads to the following equation:

$$D(q_l)\dot{r}_l + C(q_l,\dot{q}_l)r_l + K_{lD}r_l = e_t \quad (7)$$

2.2. Joint torque controller

In order to drive the actual joint torque to follow the desired joint torque τ_{sd}, a computed-torque like control law is given as:

$$u = I_{ts}\dot{v}_t + B_{ts}v_t + h_{ts}(q_l,\dot{q}_l,\tau_s) - K_{tD}(\dot{\tau}_s - v_t) \quad (8)$$

Then, from equation (5), we have:

$$I_{ts}\dot{r}_t + B_{ts}r_t + K_{tD}r_t = 0 \quad (9)$$

where $v_t = \dot{\tau}_{sd} - \Lambda_t(\tau_s - \tau_{sd}) = \dot{\tau}_{sd} - \Lambda_t e_t$, $r_t = \dot{\tau}_s - v_t$, and K_{tD} and Λ_t are diagonal positive definite gain matrices.

Convergence Analysis

The asymptotic stability and convergence properties of the entire control system is analyzed based on Lyapunov stability theory. First, an important lemma is introduced and then theorem 1 is proved as follows.

Lemma 1 *[15] Suppose that a symmetric matrix Q is partitioned as:*

$$Q = \begin{bmatrix} Q_{11} & Q_{12} \\ Q_{12}^T & Q_{22} \end{bmatrix}$$

where Q_{11} and Q_{22} are square. The matrix Q is positive definite if and only if Q_{11} is positive definite and $Q_{22} > Q_{12}^T Q_{11}^{-1} Q_{12}$.

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Theorem 1 *Consider the robot dynamic system (1) and (5) satisfying assumptions A1 to A3. The motion controller (6) and joint torque controller (8) stabilize the system and achieve global asymptotic convergence, i.e.*

$$\lim_{t\to\infty} r_l = 0, \qquad \lim_{t\to\infty} r_t = 0$$

Proof: Substituting the control law (6) and (8) into robot dynamics equations (1) and (5), yields two sets of dynamic error equations (7) and (9). Choosing the Lyapunov function candidate as

$$V = \frac{1}{2} r_l^T D(q_l) r_l + \frac{1}{2} r_t^T I_{ts} r_t + e_t^T \Lambda_t^T K_{tD} e_t \; > 0 \quad (10)$$

the differentiation of V is

$$\begin{aligned} \dot{V} &= r_l^T D(q_l)\dot{r}_l + \frac{1}{2} r_l^T \dot{D}(q_l) r_l + r_t^T I_{ts} \dot{r}_t + 2 e_t^T \Lambda_t^T K_{tD} \dot{e}_t \\ &= r_l^T D(q_l)\dot{r}_l + \frac{1}{2} r_l^T \dot{D}(q_l) r_l - r_t^T B_{ts} r_t - \dot{e}_t^T K_{tD} \dot{e}_t \\ &\quad - e_t^T \Lambda_t^T K_{tD} \Lambda_t e_t \end{aligned} \quad (11)$$

From equation (7) and the fact that $(\dot{D} - 2C)$ is skew symmetric, we have

$$\begin{aligned} \dot{V} &= -r_l^T K_{lD} r_l + r_l^T e_t - r_t^T B_{ts} r_t - \dot{e}_t^T K_{tD} \dot{e}_t \\ &\quad - e_t^T \Lambda_t^T K_{tD} \Lambda_t e_t \\ &= -\begin{bmatrix} r_l^T & e_t^T \end{bmatrix} Q \begin{bmatrix} r_l \\ e_t \end{bmatrix} - r_t^T B_{ts} r_t - \dot{e}_t^T K_{tD} \dot{e}_t \end{aligned} \quad (12)$$

with

$$Q = \begin{bmatrix} K_{lD} & -\frac{1}{2} I \\ -\frac{1}{2} I & \Lambda_t^T K_{tD} \Lambda_t \end{bmatrix}$$

By lemma 1, for matrix Q to be a positive definite matrix, the requirement is

$$\Lambda_t^T K_{tD} \Lambda_t > \frac{1}{4} K_{lD}^{-1}$$

Since K_{tD}, K_{lD} and Λ_t are diagonal matrices of controller gains, we can choose them properly to meet the requirement, such that $Q > 0$, and $\dot{V} < 0$.

Thus, the system is globally asymptotically stable in the sense of Lyapunov. r_l and r_t are bounded and belong to L_2. Also, from the equations (7) and (9), $\dot{r}_l$ and $\dot{r}_t$ are bounded as well. That means r_l and r_t are uniformly continuous functions, and so is $\dot{V}$. Therefore,

$$\lim_{t\to\infty} r_l = 0, \qquad \lim_{t\to\infty} r_t = 0$$

i.e. $e_l \to 0$, $\dot{e}_l \to 0$ and $e_t \to 0$ $\dot{e}_t \to 0$ as $t \to \infty$. This completes the proof of the theorem.

Implementation Issues

According to the control law, we notice that to generate the desired joint torque τ_{sd}, we only need the signals $q_l, \dot{q}_l$ and the desired trajectory q_d up to the second derivative, while the synthesis of input control torque u requires the first and second order derivations of τ_{sd}. From (6), we have

$$\begin{aligned} \dot{\tau}_{sd} &= D(q_l)\ddot{v}_l + C(q_l, \dot{q}_l)\dot{v}_l + \dot{D}(q_l)\dot{v}_l + \dot{C}(q_l, \dot{q}_l) v_l \\ &\quad + \dot{G}(q_l) - K_{lD}(\ddot{q}_l - \dot{v}_l) \end{aligned} \quad (13)$$

This requires information on $q_d^{(3)}$ and $\ddot{q}_l$ in addition. But from the dynamic equation (1), $\ddot{q}_l$ can be calculated by τ_s, q_l and $\dot{q}_l$, namely,

$$\begin{aligned} \ddot{q}_l &= D^{-1}(\tau_s - C(q_l, \dot{q}_l)\dot{q}_l - G(q_l)) \\ &= f_1(q_l, \dot{q}_l, \tau_s) \end{aligned}$$

Substituting it into (13), we have

$$\dot{\tau}_{sd} = f_2(q_d, \dot{q}_d, \ddot{q}_d, q_d^{(3)}, q_l, \dot{q}_l, \tau_s) \quad (14)$$

Derivating it again, we have

$$\begin{aligned} \ddot{\tau}_{sd} &= f_3(q_d, \dot{q}_d, \ddot{q}_d, q_d^{(3)}, q_d^{(4)}, q_l, \dot{q}_l, \ddot{q}_l, \tau_s, \dot{\tau}_s) \\ &= f_4(q_d, \dot{q}_d, \ddot{q}_d, q_d^{(3)}, q_d^{(4)}, q_l, \dot{q}_l, \tau_s, \dot{\tau}_s) \end{aligned} \quad (15)$$

thus, according to equation (8),

$$u = f_5(q_d, \dot{q}_d, \ddot{q}_d, q_d^{(3)}, q_d^{(4)}, q_l, \dot{q}_l, \tau_s, \dot{\tau}_s) \quad (16)$$

which means that if the assumptions A1 to A3 are satisfied, the control method can be implemented.

In practice, $\dot{\tau}_s$ is not measured directly but is approximated by the first backward difference. Since the sampling rate is high, this approximation is accurate enough for motion control purposes.

3 Robust adaptive control with all parameters unknown

In practice, there are uncertainties in the manipulator parameters. Here, we assume that all parameters, both in link dynamics and drive system, including the joint flexibility values, are unknown except for some of their bounds. Based on the result of the previous section, a robust adaptive control algorithm is developed as follows.

3.1. Adaptive motion controller

According to the fact P2, we can rewrite equation (1) as [14]

$$D(q_l)\ddot{q}_l + C(q_l, \dot{q}_l)\dot{q}_l + G(q_l) = Y(q_l, \dot{q}_l, \ddot{q}_l) P = \tau_s \quad (17)$$

where $Y(.)$ is an $n \times r$ matrix of known functions, referred to as the regressor, and P is an r-dimensional vector of parameters.

The adaptive control law is

$$\begin{aligned} \tau_{sd} &= \hat{D}(q_l)\dot{v}_l + \hat{C}(q_l, \dot{q}_l) v_l + \hat{G}(q_l) - K_{lD}(\dot{q}_l - v_l) \\ &= Y(q_l, \dot{q}_l, v_l, \dot{v}_l)\hat{P} - K_{lD} r_l \end{aligned} \quad (18)$$

where $\hat{D}, \hat{C}, \hat{G}$ and $\hat{P}$ are estimated parameters. Substituting (18) into equation (1), gives

$$D(q_l)\dot{r}_l + C(q_l, \dot{q}_l) r_l + K_{lD} r_l = Y(q_l, \dot{q}_l, v_l, \dot{v}_l)\tilde{P} + e_t \quad (19)$$

with $\tilde{P} = \hat{P} - P$

3.2. Robust joint torque controller

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We note that h_{ts} in equation (5) contains the mixed uncertainties of link dynamics and drive system. It would make the parameter update (if using the adaptive control method) or estimation of the uncertainty bounds (if using a robust control) somewhat difficult. However, by assuming that $\ddot{q}_l$ is available for the control system, we can write equation (5) as follows:

$$I_{ts}\ddot{\tau}_s + B_{ts}\dot{\tau}_s + I_m\ddot{q}_l + B_m\dot{q}_l + \tau_s = u \qquad (20)$$

thus h_{ts} is equal to $I_m\ddot{q}_l + B_m\dot{q}_l + \tau_s$. Defining a $n \times p$ matrix Y_t and a p-dimensional vector P_t as

$$Y_t(\dot{q}_l, \ddot{q}_l, \tau_s, \dot{\tau}_s, \ddot{\tau}_s) = [diag(\ddot{\tau}_s), diag(\dot{\tau}_s), diag(\ddot{q}_l), diag(\dot{q}_l), \tau_s]$$

$$P_t = [..I_{tsi}.., ..B_{tsi}.., ..I_{mi}.., ..B_{mi}.., 1]^T$$

The equation (20) becomes

$$Y_t(\dot{q}_l, \ddot{q}_l, \tau_s, \dot{\tau}_s, \ddot{\tau}_s)P_t = u$$

We assume that the uncertainty of the parameters is bounded, that is, for the available parameters $\hat{P}_t$, there exists $\rho \in R_+$, such that,

$$\| \tilde{P}_t \| = \| \hat{P}_t - P_t \| \leq \rho \qquad (21)$$

Based on (8), using control law:

$$u = \hat{I}_{ts}\dot{v}_t + \hat{B}_{ts}v_t + \hat{h}_{ts} - K_{tD}(\dot{\tau}_s - v_t) + Y_t\Delta u \qquad (22)$$

we have

$$I_{ts}\dot{r}_t + B_{ts}r_t + K_{tD}r_t = (\hat{I}_{ts} - I_{ts})\dot{v}_t + (\hat{B}_{ts} - B_{ts})v_t + (\hat{h}_{ts} - h_{ts}) + Y_t\Delta u = Y_t(\dot{q}_l, \ddot{q}_l, \tau_s, v_t, \dot{v}_t)(\tilde{P}_t + \Delta u) \qquad (23)$$

where Δu is defined in (24).

Theorem 2 *Using the controller (18) and (22) with*

$$\Delta u = \begin{cases} -\rho\dfrac{Y_t^T r_t}{\| Y_t^T r_t \|} & \text{if } \| Y_t^T r_t \| > \epsilon \\ -\dfrac{\rho}{\epsilon}Y_t^T r_t & \text{if } \| Y_t^T r_t \| \leq \epsilon \end{cases} \qquad (24)$$

and parameter update law

$$\dot{\hat{P}} = -\Gamma^{-1}Y^T(q_l, \dot{q}_l, v_l, \dot{v}_l)r_l \qquad (25)$$

the system is uniformly stable and the tracking error is uniformly ultimately bounded (u.u.b.).

Proof: Choosing the Lyapunov function candidate as

$$V = \frac{1}{2}r_l^T D(q_l)r_l + \frac{1}{2}\tilde{P}^T\Gamma\tilde{P} + \frac{1}{2}r_t^T I_{ts}r_t + e_t^T\Lambda_t^T K_{tD}e_t > 0 \qquad (26)$$

through a similar calculation of previous section, the differentiation of V is

$$\dot{V} = -\begin{bmatrix} r_l^T & e_t^T \end{bmatrix} Q \begin{bmatrix} r_l \\ e_t \end{bmatrix} - r_t^T B_{ts} r_t - \dot{e}_t^T K_{tD}\dot{e}_t + \tilde{P}^T(Y^T r_l + \Gamma\dot{\hat{P}}) + (Y_t^T r_t)^T(\tilde{P}_t + \Delta u) \qquad (27)$$

With the parameter update law (25) and control Δu (24) we can show that $\dot{V} < 0$. Let us examine the last term in the above. If $\| Y_t^T r_t \| > \epsilon$, then

$$(Y_t^T r_t)^T(\tilde{P}_t + \Delta u) = (Y_t^T r_t)^T(\tilde{P}_t - \rho\frac{Y_t^T r_t}{\| Y_t^T r_t \|}) \leq \| Y_t^T r_t \| (\| \tilde{P}_t \| - \rho) < 0 \qquad (28)$$

If $\| Y_t^T r_t \| \leq \epsilon$,

$$(Y_t^T r_t)^T(\tilde{P}_t + \Delta u) \leq (Y_t^T r_t)^T(\rho\frac{Y_t^T r_t}{\| Y_t^T r_t \|} + \Delta u) = (Y_t^T r_t)^T(\rho\frac{Y_t^T r_t}{\| Y_t^T r_t \|} - \frac{\rho}{\epsilon}Y_t^T r_t) \qquad (29)$$

This last term achieves a maximum value of $\epsilon\rho/4$ when $\| Y_t^T r_t \| = \epsilon/2$. That is,

$$\dot{V} \leq -\begin{bmatrix} r_l^T & e_t^T \end{bmatrix} Q \begin{bmatrix} r_l \\ e_t \end{bmatrix} - r_t^T B_{ts} r_t - \dot{e}_t^T K_{tD}\dot{e}_t + \frac{\epsilon\rho}{4} \leq -\lambda_{min}(Q) \| (r_l^T, e_t^T) \|^2 - min(B_{tsi}) \| r_t \|^2 - min(K_{tDi}) \| \dot{e}_t \|^2 + \frac{\epsilon\rho}{4} \qquad (30)$$

If $\lambda_{min}(Q) \| (r_l^T, e_t^T) \|^2 + min(B_{tsi}) \| r_t \|^2 + min(K_{tDi}) \| \dot{e}_t \|^2 > \epsilon\rho/4$, then $\dot{V} < 0$, uniform stability and uniform ultimate boundedness follow using the results of Leitmann [16].

Remark 1: It is noticed that using a single bound ρ to measure the parameter uncertainty may lead to overly conservative design and limited the adjustable capability of the controller [17]. It is easy to partition $Y_t\Delta u$ in our robust controller in case of need.

Remark 2: In implementation, it should be kept in mind the fact that the estimated parameters may not converge to their true values by Slotine and Li method, and the difference in the control mechanisms of the rigid and flexible joint robots. In the rigid case, the estimated parameters generate the control input u directly. In the flexible joint case, the estimated parameter $\hat{P}$ generates the desired torque command τ_{sd}. The divergence of $\hat{P}$ from the true value P is reflected in the variation of τ_{sd}, and it may cause large changes in $\dot{\tau}_{sd}$ and $\ddot{\tau}_{sd}$, and consequently a large change in input u. Since the control method is digitally implemented, these large changes may become untolerable before they are under control. Therefore, to ensure the convergence of the control system, it is better to restrict the parameter estimations within certain bounds [18], such that

$$l_i < P_i < h_i$$

The parameter update law (25) is thus augmented with the reseting conditions as follows,

$$\hat{P}_i(t^+) = \begin{cases} l_i & \text{if } \hat{P}_i(t) < l_i \\ h_i & \text{if } \hat{P}_i(t) > h_i \\ \hat{P}_i(t) & \text{otherwise} \end{cases} \qquad (31)$$

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In (26), let's define r_t and e_t at time t^+. Similar to [18], it can be shown that the addition of parameter reseting maintains the negative of $\dot{V}$, i.e. the theorem 2 remains valid.

4 Simulation examples

To verify the effectiveness of the control methods proposed in this paper, numerical simulations are carried out based on the dynamic model of the IRIS robot established in the Robotics and Automation Laboratory at the University of Toronto [12]. Consider the motion control of the robot manipulator with an upper arm and a forearm located in a vertical plane, as shown in Figure 1:

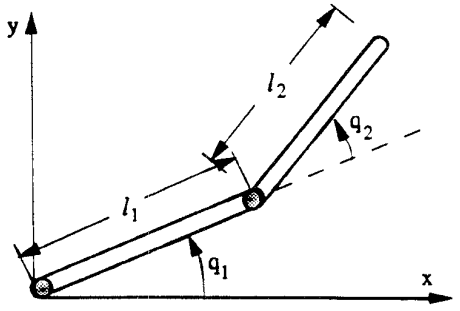


Figure 1: The equivalent diagram of a two DOF robot manipulator

The dynamic model of the rotor subsystems with joint torque sensors is given as follows:

$$\gamma_1(\gamma_1+1)J_{m1}\ddot{q}_{m1}+\gamma_1(\gamma_1+1)b_{m1}\dot{q}_{m1}+\tau_{s1}=u_1$$
$$\gamma_2(\gamma_2+1)J_{m2}\ddot{q}_{m2}+\gamma_2(\gamma_2+1)b_{m2}\dot{q}_{m2}+\tau_{s2}=u_2$$

and the corresponding link dynamics is:

$$H_{11}\ddot{q}_{l1}+H_{12}\ddot{q}_{l2}-h\dot{q}_{l2}^2-2h\dot{q}_{l1}\dot{q}_{l2}+G_1=\tau_{s1}$$
$$H_{21}\ddot{q}_{l1}+H_{22}\ddot{q}_{l2}+h\dot{q}_{l1}^2+G_2=\tau_{s2}$$

The simulations are performed using Simnon. The control schemes are digitally implemented with the sampling rate of 1msec. The desired trajectories are generated by fifth-order polynomials. Joint 1 moves from 90 to 0 (degree), joint 2 moves from 0 to 60 (degree), both in the first second, then hold constant afterwards. The responses of the systems are shown in Figure 2 and 3. The tracking errors of the system with known parameters and the robust adaptive control system converge to zeros. Their remarkable efficiency is verified. The parameter estimation curves are plotted in Figure 3 as well. It can be seen that some estimated parameters do not converge to their true values, and the reseting is necessary.

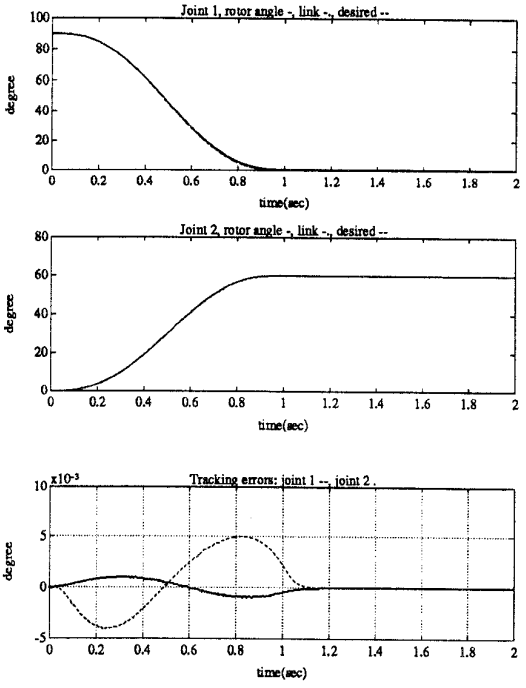


Figure 2: The response of the system with known parameters

5 Conclusion

In this paper, motion control of flexible joint robots with joint torque feedback is considered. A two-stage control scheme consisting of a motion controller and a joint torque controller is established in a systematic way for the general n-link case. To deal with uncertainties in the robotic system, a robust adaptive control algorithm is developed assuming that all the system parameters including the joint flexibility values are unknown except for some of their bounds. The result compared with the previous work has the distinct feature that no restriction of joint flexibility, nor exact knowledge of the parameters of rotor subsystem is required. The system stability is analyzed via Lyapunov stability theory. Simulation results illustrate the effectiveness of the proposed control scheme.

It is noticed that with the joint torque controller, the methods developed for rigid robots can be used for the control design of flexible joint robots. In our control scheme, the motion controller is designed by the method of Slotine and Li's; the robust joint torque controller is developed in the similar way. It is obvious that other control methods, such as computed-torque control, can be also incorporated in the control scheme. Hence, the control scheme developed in this paper provides a useful framework for the control of flexible joint robots.

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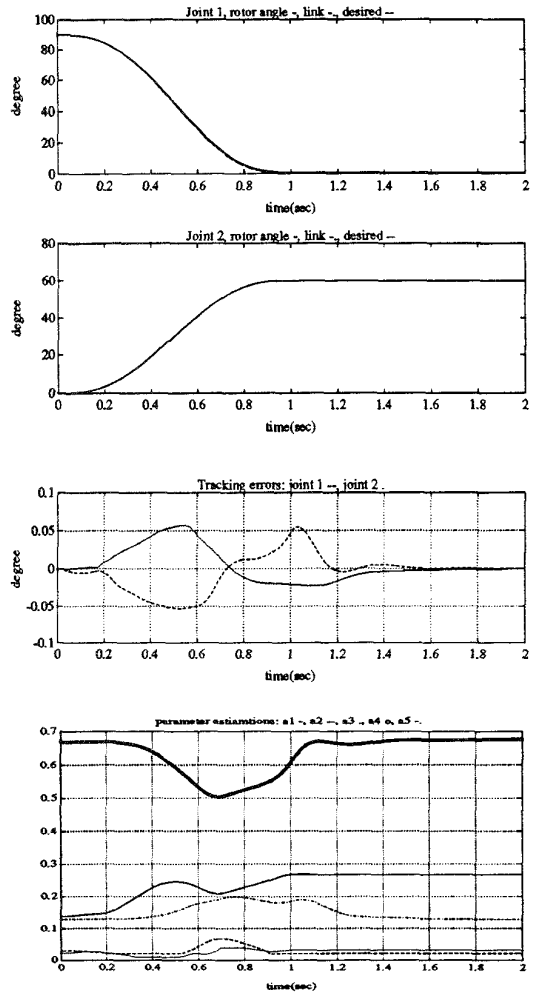


Figure 3: The response of the robust adaptive control system

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